TOPIC 02 · SKILLS AND RULES
LAB 6 THEME · CUSTOM SKILLS: PARETO CHART

Lab 6 — Use Copilot to Build a Custom Pareto-Chart Skill

Maps to	LO2: use Copilot on the web to write a custom SKILL.md, upload it to the OneDrive skills folder, install it in Excel and call it with @ to plot a Pareto chart from a one-line prompt
Duration	45 minutes
Tools	Copilot on the web (copilot.microsoft.com) to draft the skill · OneDrive skills folder · Excel Copilot (Chat only, Allow editing) · SKILL.md
Mock data	data/BrightPath_Job_Postings.xlsx
Lab files	skills/pareto-chart/SKILL.md (ready-made skill to upload if you run short of time)
Course	Microsoft 365 Copilot for Excel (C34) · v3.1
Prompts	https://tertiarycourses.github.io/C34-Microsoft-365-Copilot-for-Excel/prompts/#lab-6

Goal

Create a pareto-chart skill that turns any category column into a Pareto chart - a different colour for each bar and a cumulative % line - and use it to plot BrightPath's 500 job postings by job category.

What You Will Do

BrightPath received 500 job postings from clients in H1 2026, spread over 16 job categories. Which few categories make up most of the work? A Pareto chart answers that: bars sorted from highest to lowest and a cumulative % line that shows where the 'vital few' end. Excel's built-in Pareto chart paints every bar the same colour, so you will write a custom skill instead. You will ask Copilot on the web to write the pareto-chart SKILL.md, save it, upload it to your OneDrive skills folder, install it in Excel and call it with @ to plot the job categories - then reuse it on another column and improve it.

What You Will Build

A pareto-chart skill in your OneDrive skills folder, and BrightPath_Job_Postings.xlsx with a Pareto_Job_Category sheet and a Pareto_Client_Sector sheet - each a summary table and a Pareto chart with a different colour for every bar - plus a Checks sheet

Prerequisites

- Lab 5 completed: you can find, enable and @mention skills.
- Access to https://copilot.microsoft.com with the account you use in Excel and OneDrive, and to your OneDrive in a browser (custom skills are stored there).
- This lab folder's data workbook and skills/pareto-chart/SKILL.md (fallback).

Data note

Use only the supplied fictional BrightPath data or information you are authorised to process. Do not paste credentials, confidential business material or real employees' personal data into Copilot, rules or skills.

Steps

Step 1. Copy BrightPath_Job_Postings.xlsx from this lab's data folder into a work-c34 folder on your computer and open it in Excel for Microsoft 365 (desktop). Work on this local copy - no OneDrive upload is needed. Check that Formulas > Calculation Options is Automatic, then open Copilot from the Copilot button on the Home tab or the floating Copilot icon at the bottom right. Read the About_This_Data sheet before you prompt. The workbook stays local; only the skill file goes in OneDrive (Steps 4-5).

PROMPT / CHECK — copy and adapt

```
Table : tblJobs (500 job postings, Jan-Jun 2026)
Columns : Job_ID, Posted_Date, Job_Category (16 values), Client_Sector (6 values),
Employment_Type, Status, Openings
Pareto : columns = jobs per category, highest to lowest, a different colour for each column;
         line = cumulative % of all jobs, 0% to 100% on a secondary axis
```

Step 2. Draft the skill in Copilot on the web. Open https://copilot.microsoft.com in your browser and sign in with the same account you use in Excel and OneDrive (with a work account you may be sent to Microsoft 365 Copilot Chat - use that chat). Start a new chat and paste the prompt below. The web chat cannot see your workbook, so the prompt describes the table, the chart and the colours. Read the reply: it must start with --- and include name: pareto-chart, a description that says when to use the skill, and numbered steps that give every column a different colour. Ask for corrections in this chat until you are happy.

PROMPT / CHECK — copy and adapt

```
PROMPT (Microsoft Copilot web):
Write a complete SKILL.md for a Copilot in Excel skill named pareto-chart.
Return reusable skill instructions only; do not draw a chart or claim to see my workbook in
this chat.
Start with YAML front matter containing name: pareto-chart and a description.
Description: Use when I ask for a Pareto chart, an 80/20 analysis or the vital few categories
in a table column.
When the skill runs in Excel I will name a table and a category column, for example tblJobs
and Job_Category. If I do not, it must ask me.
Steps:
1) Never change the source table or any existing sheet.
2) Create a new sheet named Pareto_<column name> with a summary table: Category, Count, Share
%, Cumulative %, sorted by Count from highest to lowest. Use formulas that read the source
table so it updates when rows are added, and add a Total row.
3) Create one combo chart from the summary: Count as columns and Cumulative % as a line with
markers on a secondary axis from 0% to 100%.
4) Give every column a different colour. Never use one colour for all the columns. Make the
line dark grey.
5) Format the chart cleanly: white background, no border or effects, data labels on the
columns, axis titles, no legend, and a title that states the finding, for example "6 of 16
categories make up 83% of the total".
6) Place the chart to the right of the summary table, starting at column G.
7) Under the table, list the vital few: the categories needed to reach 80% and their combined
share.
8) Check that the Count total equals the number of rows in the source table, and report the
row count, the number of categories, the top category and how many categories reach 80%.
Return only the full SKILL.md text, starting with the --- front matter, without code fences or
commentary.
```

Step 3. Save the reply as a plain-text file named SKILL.md. Select the whole reply and copy it - if it sits in a code block, copy only the text inside, with no ``` lines. Windows: open Notepad, paste, choose File > Save as, set Save as type to All files, type SKILL.md as the file name, set Encoding to UTF-8 and save it in work-c34. Mac: open TextEdit, choose Format > Make Plain Text, paste, choose File > Save, type SKILL.md as the name, untick 'If no extension is provided, use .txt' and save it in work-c34. Check the file is called SKILL.md, not SKILL.md.txt, and that its first line is ---.

PROMPT / CHECK — copy and adapt

```
File : work-c34/SKILL.md (plain text)
Line 1 : ---
Line 2 : name: pareto-chart
Check : no ``` lines and no text before the first ---
Short of time? Use skills/pareto-chart/SKILL.md from this lab folder instead.
```

Step 4. Open your OneDrive skills folder from Excel. In Excel open the Copilot pane and select Settings (...) > Manage plugins & skills > Custom skills. Turn the Custom skills switch on, then select Add in OneDrive (in some versions: Create OneDrive folder, then Open skills folder). Your browser opens OneDrive at the skills folder, usually Documents > Copilot > Microsoft Excel > skills. A sample folder such as make-calendar.example shows the layout: one folder per skill.

PROMPT / CHECK — copy and adapt

```
Excel : Copilot pane > Settings (...) > Manage plugins & skills > Custom skills
Switch : Custom skills = On
Then : Add in OneDrive (or Create OneDrive folder, then Open skills folder)
OneDrive : Documents > Copilot > Microsoft Excel > skills
```

Step 5. Create the skill's folder and upload SKILL.md. In the OneDrive skills folder select Create or upload > Folder, name it pareto-chart (lower case, with a hyphen) and select Create. Open the new folder, select Create or upload > Files upload, choose SKILL.md from work-c34 and select Open. When the upload finishes, SKILL.md is listed inside the pareto-chart folder.

PROMPT / CHECK — copy and adapt

```
OneDrive : Documents > Copilot > Microsoft Excel > skills > pareto-chart > SKILL.md
Check : folder name = the name: line = pareto-chart (lower case, hyphen, no spaces)
Check : the file is called SKILL.md and sits directly inside the pareto-chart folder
```

Step 6. Install the skill in Excel. Go back to Excel: Copilot pane > Settings (...) > Manage plugins & skills > Custom skills. Check the Custom skills switch is on and select Refresh (the round arrow). Wait until the list loads, then turn pareto-chart on. Close the dialog, click the Copilot prompt box and type @pare - pareto-chart should appear in the list. If it does not, wait a minute and select Refresh again.

PROMPT / CHECK — copy and adapt

```
Excel : Copilot pane > Settings (...) > Manage plugins & skills > Custom skills > Refresh
Then : turn pareto-chart on, close the dialog and type @pare in the prompt box
Check : pareto-chart appears in the list
```

Step 7. Use the skill. First check that the title bar shows BrightPath_Job_Postings: a skill is available in every workbook, but it only works where its table exists. Choose Allow editing and send the one-line prompt. Copilot should create a sheet Pareto_Job_Category with a summary table and a combo chart beside it: columns from Customer Service (145) down to Procurement (1), a different colour for each column, and a cumulative % line that passes 80% at category 6. Inspect it and select Done. Then ask Copilot to add a Checks sheet so you can compare the numbers.

PROMPT / CHECK — copy and adapt

```
ALLOW EDITING:
@pareto-chart Plot a Pareto chart of Job_Category in tblJobs.

CHECK (Allow editing):
Add a Checks sheet with formulas that count all the jobs, the Customer Service jobs and the
number of job categories, so I can check the Pareto chart.
```

Step 8. Reuse it. This is the point of a skill: a different column needs only its name. Run it on Client_Sector, then ask Copilot which job categories make up the vital few.

PROMPT / CHECK — copy and adapt

```
ALLOW EDITING:
@pareto-chart Plot a Pareto chart of Client_Sector in tblJobs.

CHAT ONLY:
Which job categories together make up 80% of the postings? Give their combined share.
```

Step 9. Improve the skill. A skill is a living recipe: improve it whenever you correct Copilot twice. Go back to the web chat, send the prompt below, then paste your current SKILL.md under it. Save the reply as SKILL.md as in Step 3. In OneDrive open the pareto-chart folder, select Create or upload > Files upload, choose the new SKILL.md and select Replace when OneDrive asks. In Excel select Custom skills > Refresh and rerun the skill on Job_Category: the chart now shows a dashed 80% line and one Other column that groups the 6 categories under 2% (29 postings, 5.8%). Finally log your prompts - fill in the two web-chat rows yourself, because Excel Copilot cannot see the web chat.

PROMPT / CHECK — copy and adapt

```
PROMPT (Microsoft Copilot web):
I will paste my current pareto-chart SKILL.md below. Update it so the chart also shows a
dashed 80% reference line and groups any category with less than 2% share into one Other
column at the end.
Preserve its front matter and numbered steps. Return the full file without code fences. If I
have not pasted the file, ask me for it first.

ALLOW EDITING:
@pareto-chart Plot a Pareto chart of Job_Category in tblJobs.

LOG (Allow editing):
In Prompt_Log, leave the web-chat rows (Step 2 / Prompt A and Step 9 / Prompt F) unchanged.
For each Excel prompt I ran, fill in What Copilot did and How I verified, and set Decision to
Keep. Do not invent results for web prompts. I will tell you any row to change to Revise or
Undo.
```

Sample Prompts

Use the app labelled for each prompt (or adapt it). Name the table and columns, state the output, and always verify the result. The same prompts are in sample-prompts.txt in this lab folder.

Copy-ready prompts online:
https://tertiarycourses.github.io/C34-Microsoft-365-Copilot-for-Excel/prompts/#lab-6

#	Purpose	Prompt
1	Write a skill · Microsoft Copilot web	In the web chat at https://copilot.microsoft.com, use the complete Step 2 prompt below to draft pareto-chart. It describes the table, columns and chart because the web chat cannot see your workbook.
2	Call the skill	@pareto-chart Plot a Pareto chart of Job_Category in tblJobs.
3	Reuse	@pareto-chart Plot a Pareto chart of Client_Sector in tblJobs.
4	Check	Add a Checks sheet with formulas that count all the jobs and the Customer Service jobs, so I can check the Pareto chart.
5	Vital few	Which job categories together make up 80% of the postings? Show their combined share.
6	Explain	Explain in plain English how to read this Pareto chart: what the columns, the line and the 80% point tell me.
7	Improve the skill · Microsoft Copilot web	I will paste my current pareto-chart SKILL.md below. Update it so the chart also shows a dashed 80% reference line and groups any category with less than 2% share into one Other column. Preserve its front matter and numbered steps. Return the full file without code fences. If I have not pasted the file, ask me for it first.

Test It

Check	Expected result
Top category	Customer Service: 145 of 500 jobs (29.0%); 16 categories in all
Vital few	6 of 16 categories reach 83.0% (80% is passed at Nursing & Healthcare)
Pareto chart	Columns high to low, a different colour for each, cumulative % line 0-100% on a secondary axis
Client_Sector Pareto	Technology first (149 jobs, 29.8%); 4 sectors reach 80.8%
Improved skill	6 categories under 2% grouped into Other (29 jobs, 5.8%)
Skill	pareto-chart appears when you type @ and runs from a one-line prompt

Checkpoint

Keep the skill: it stays in your OneDrive and works in any workbook with a table and a category column. Topic 3 uses rules and skills to build dashboards.

Troubleshooting

If this happens	Fix
The skill does not appear after Refresh.	The folder name must match the name: line exactly (lower case, hyphens) and the file must be SKILL.md directly inside that folder. Select Refresh again and turn pareto-chart on.
There is no Add in OneDrive or Create OneDrive folder button.	Custom skills are rolling out and depend on your licence and update channel. Use the ready-made SKILL.md to read the format, and run the same steps as a normal prompt instead.
Copilot says tblJobs is not present in this workbook.	The skill ran in another lab's workbook. Open BrightPath_Job_Postings.xlsx and send the prompt again from its Copilot pane.
The web chat draws a chart or asks me to upload my workbook instead of writing SKILL.md.	Paste the full Step 2 prompt again. It asks for reusable skill instructions only; the workbook is used later in Excel, so nothing needs uploading.
The file saved as SKILL.md.txt, or it contains ``` lines.	Save as plain text with the exact name SKILL.md (Windows: Save as type All files; Mac: Format > Make Plain Text) and delete any ``` lines before the first ---.
All the columns are the same colour.	Ask: 'Give every column a different colour and keep the cumulative line dark grey.' Then add that sentence to the skill's steps, save, refresh and rerun.
The cumulative % line looks flat.	It is on the same axis as the columns. Ask: 'Move Cumulative % to a secondary axis from 0% to 100%.' Then add that to the skill's steps.

Challenge

Ask Copilot in Excel to create mock data of 500 rows for another question - for example helpdesk tickets with 12 issue categories, as a table named tblTickets. Run @pareto-chart on the Issue_Category column and name the vital few.

Reflection

Which report in your own job is really a Pareto question - a long list where a few causes matter most - and what would the skill's description say?